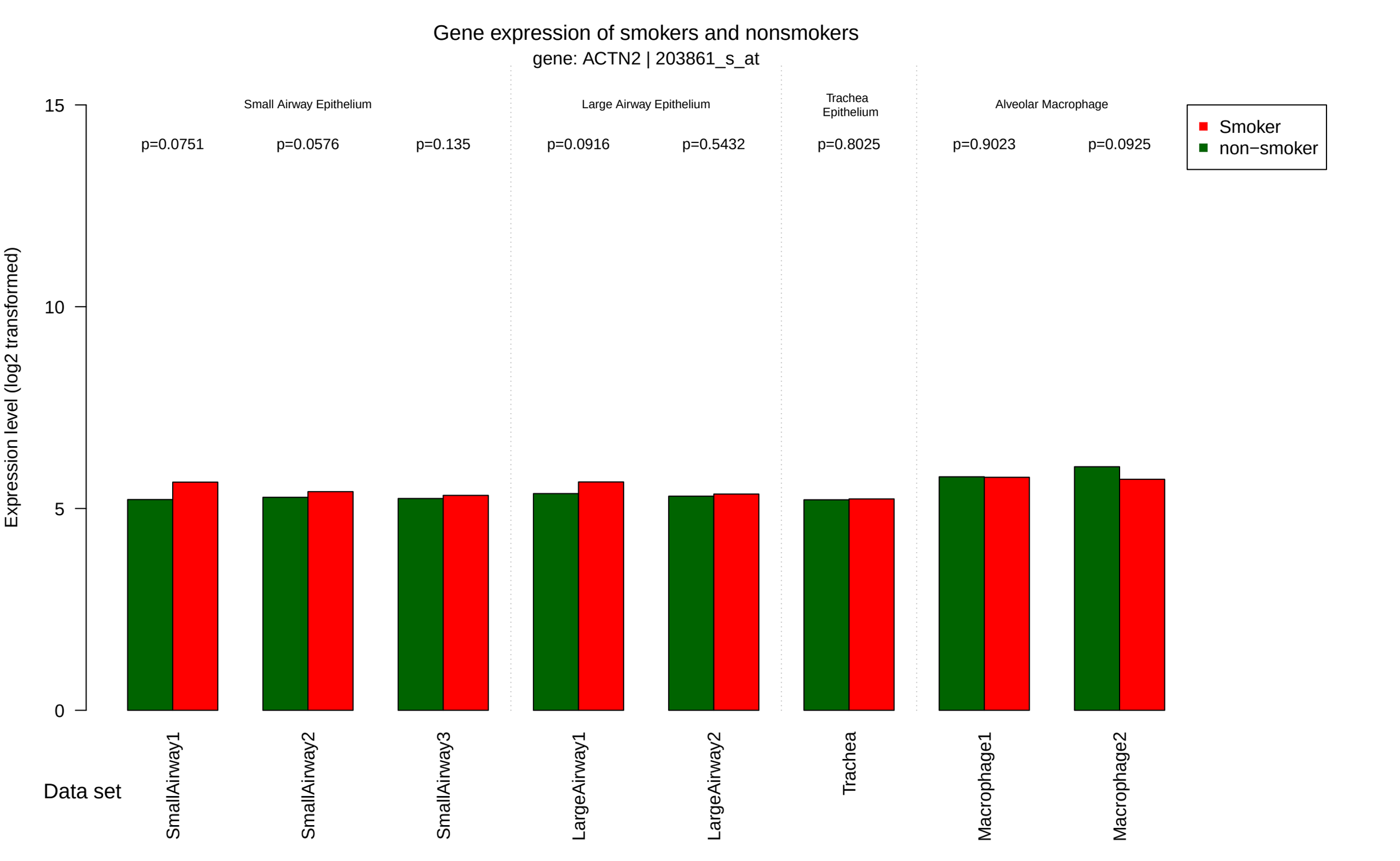
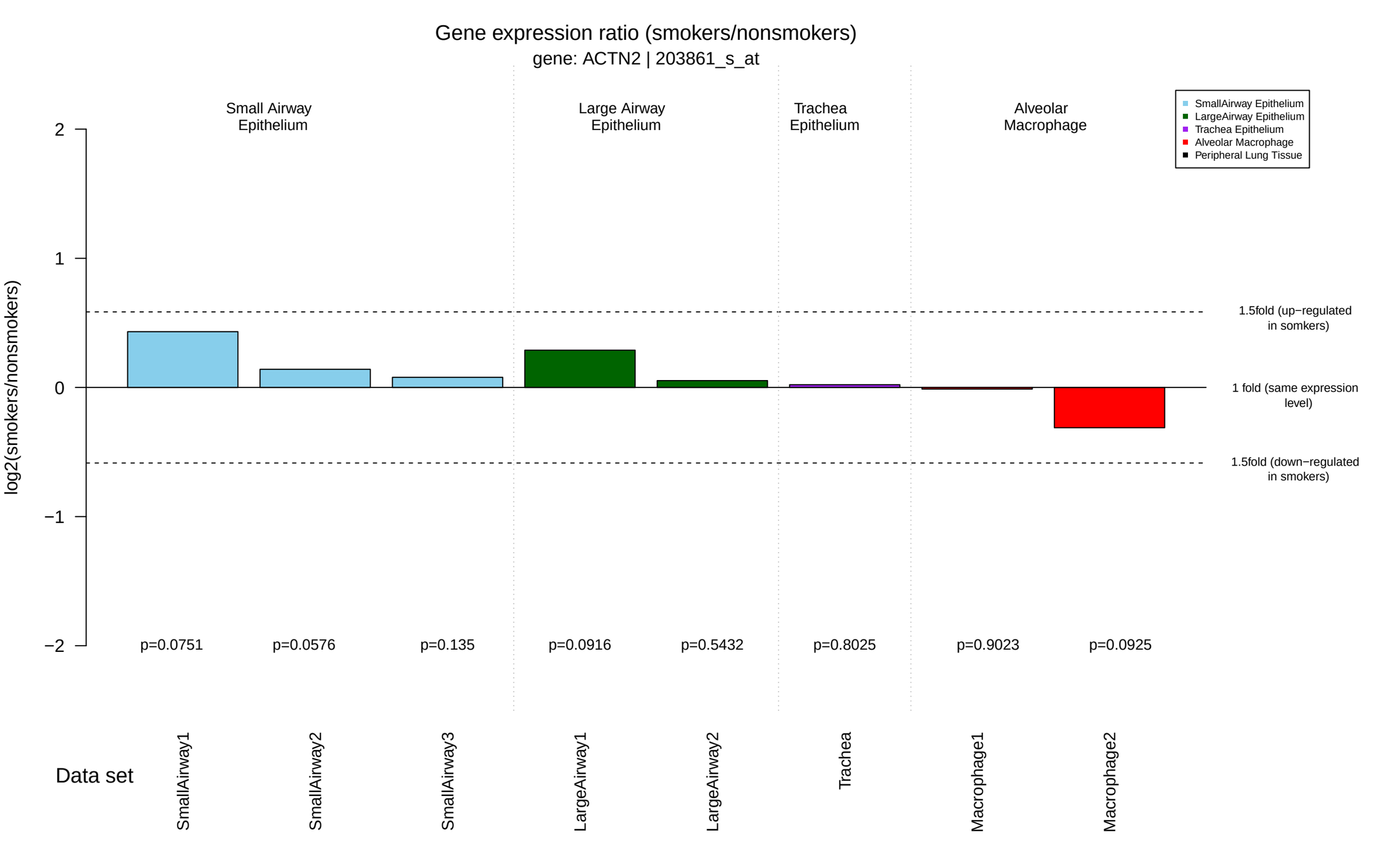
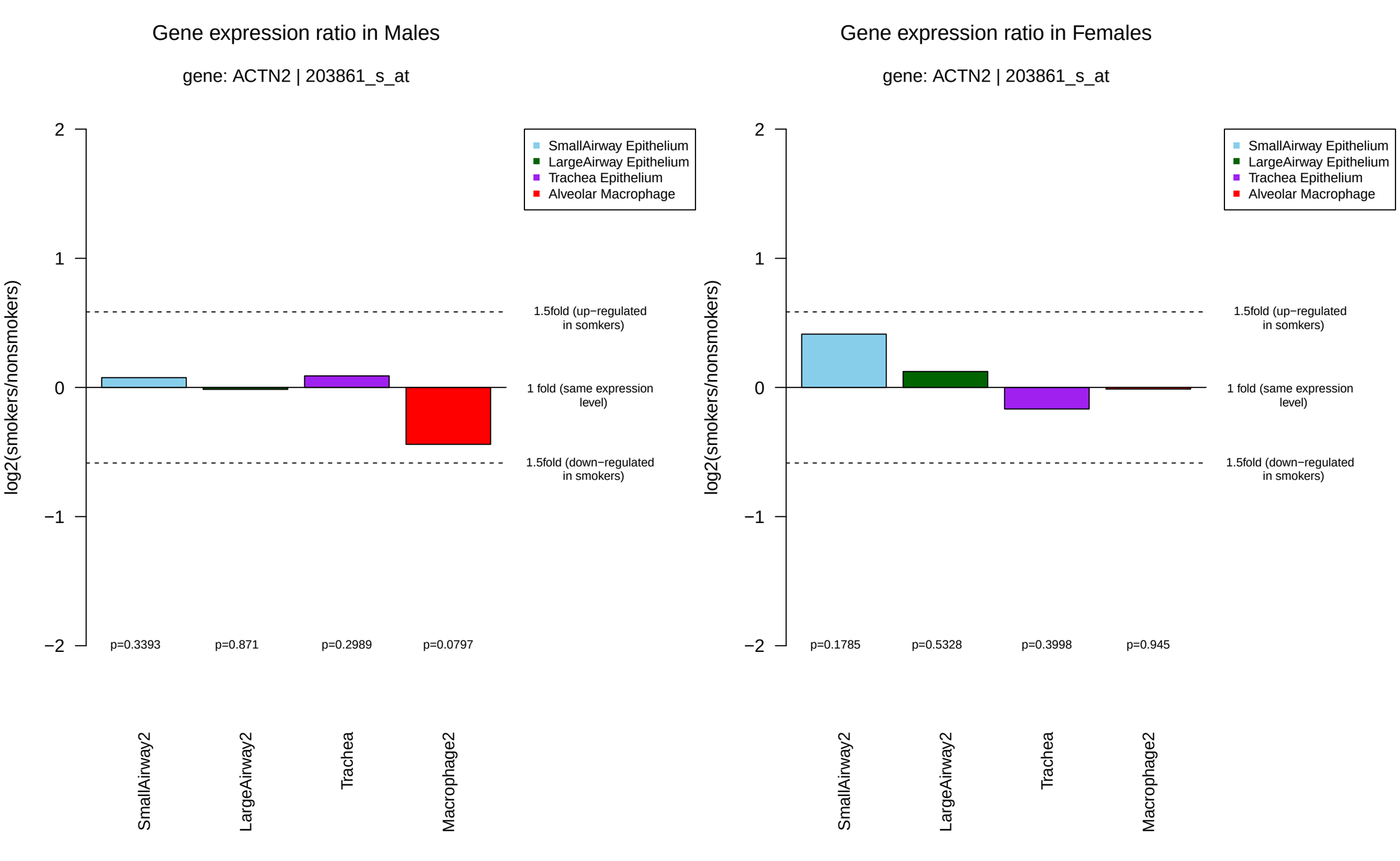
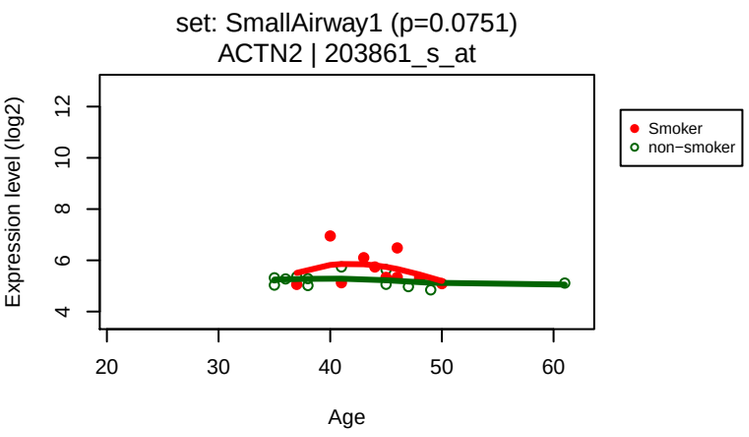
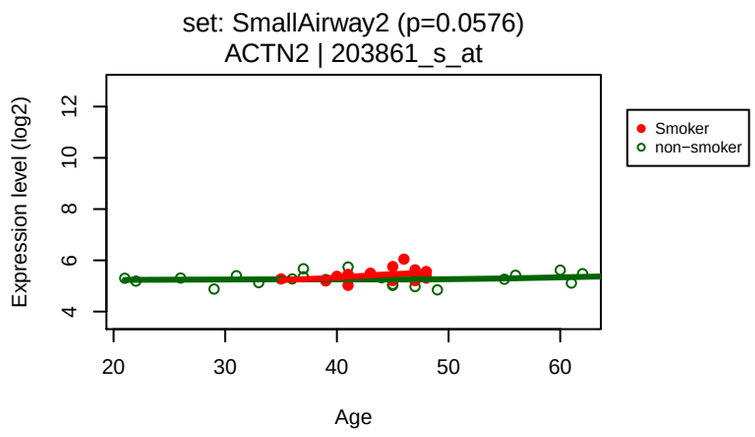
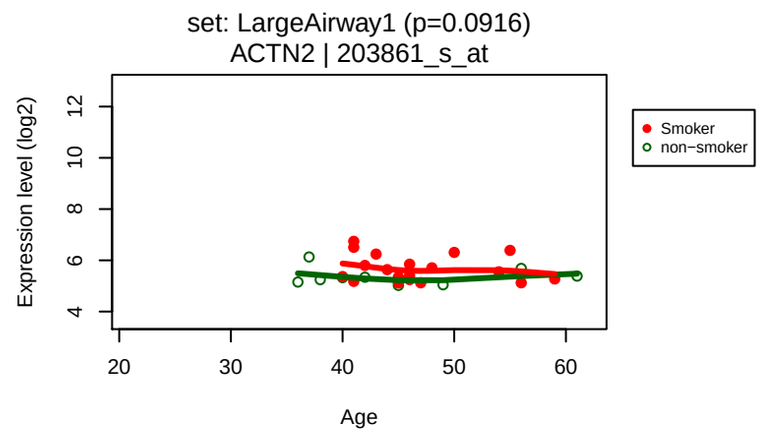

Gene expression of smokers and nonsmokers
gene: ACTN2 | 203861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0751
p=0.0576
p=0.135
p=0.0916
p=0.5432
p=0.8025
p=0.9023
p=0.0925
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

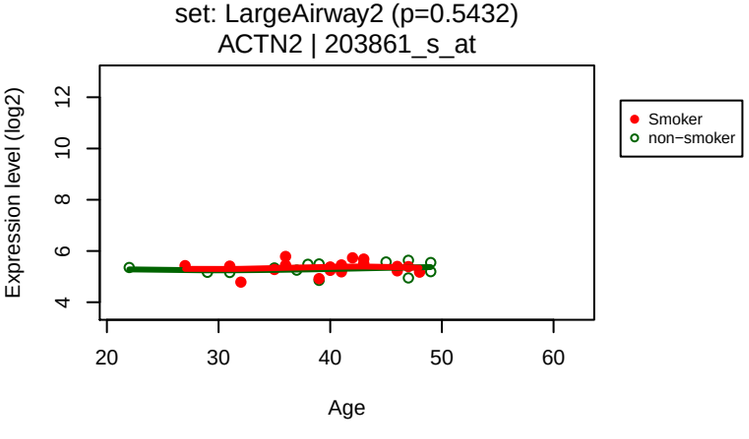

Gene expression ratio (smokers/nonsmokers)
gene: ACTN2 | 203861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0751
p=0.0576
p=0.135
p=0.0916
p=0.5432
p=0.8025
p=0.9023
p=0.0925
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

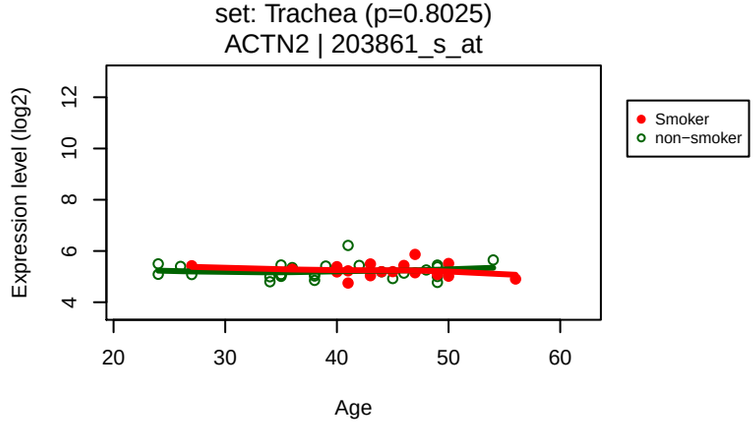

Gene expression ratio in Males
gene: ACTN2 | 203861_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3393
p=0.871
p=0.2989
p=0.0797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN2 | 203861_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1785
p=0.5328
p=0.3998
p=0.945
SmallAirway2
LargeAirway2
Trachea
Macrophage2

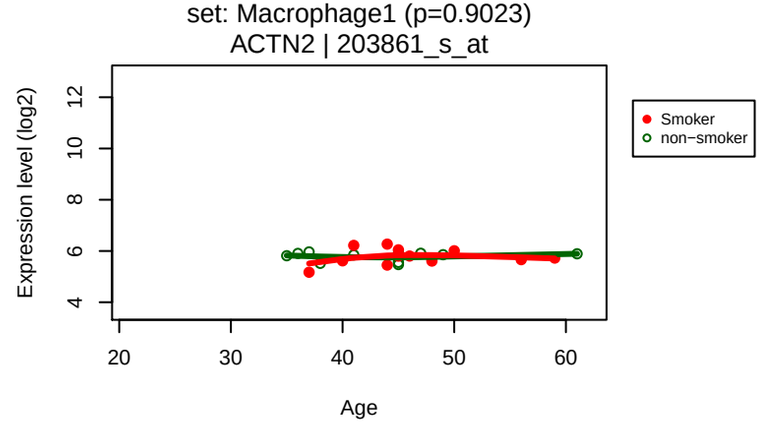

set: SmallAirway1 (p=0.0751)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker

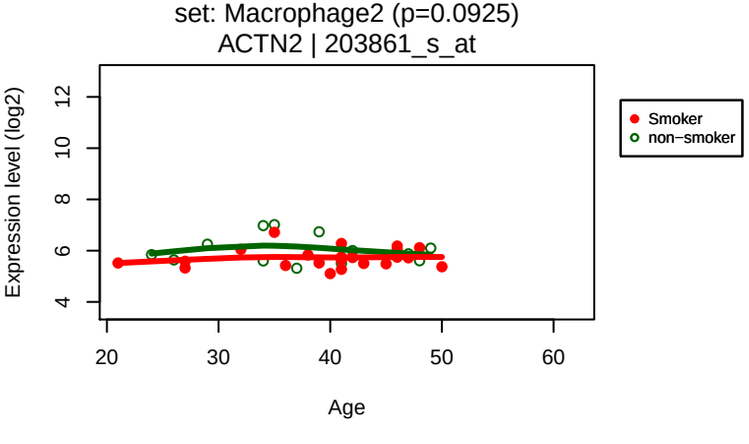

set: SmallAirway2 (p=0.0576)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker

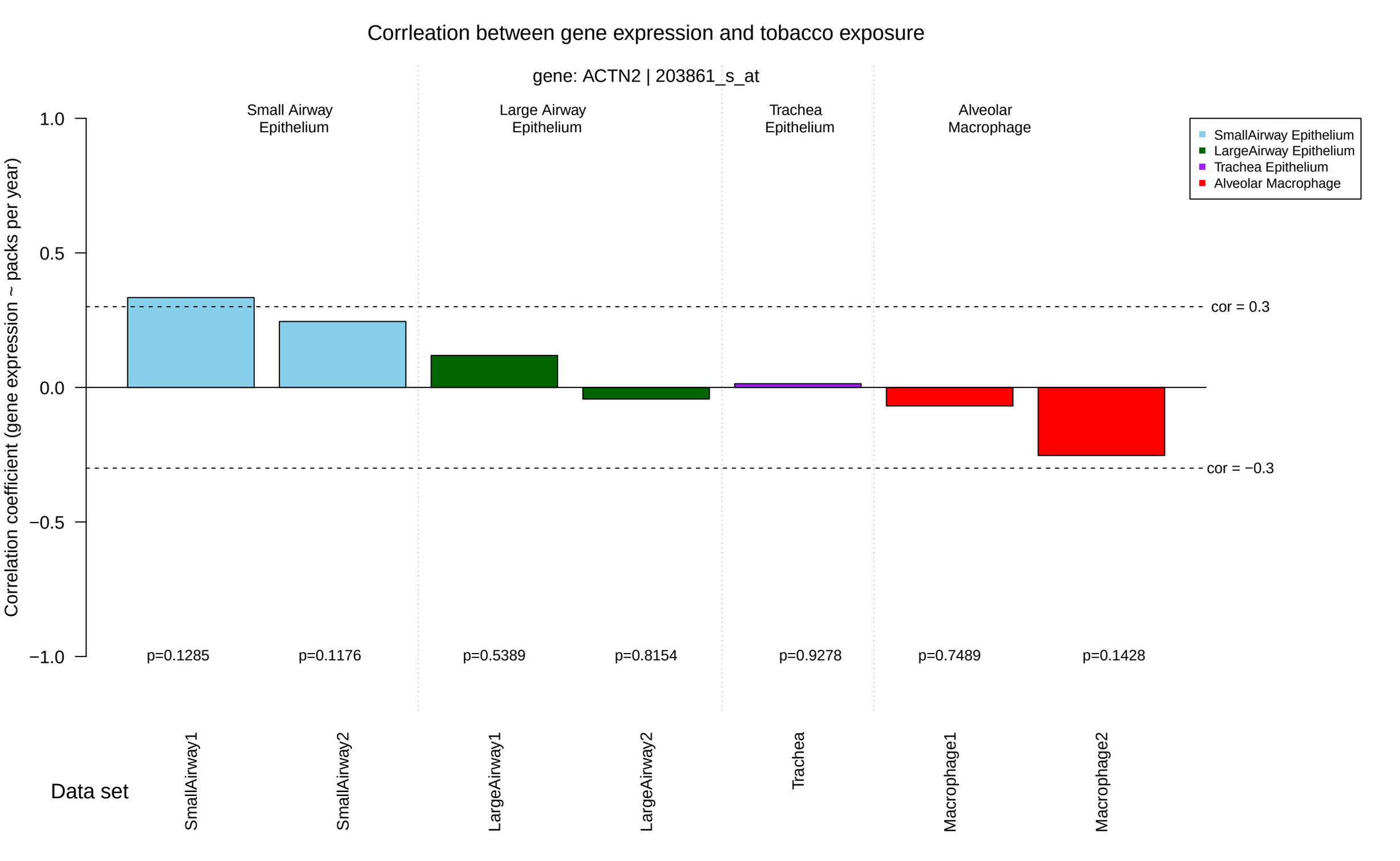

set: LargeAirway1 (p=0.0916)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5432)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8025)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9023)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0925)
ACTN2 | 203861_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN2 | 203861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1285
p=0.1176
p=0.5389
p=0.8154
p=0.9278
p=0.7489
p=0.1428
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2